AKASH SIDDAPPA GOUNDI

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EDUCATION

R V College of Engineering — B.E. in Information Science and Engineering — ***2023 – 2027*** | *CGPA:* ***8.37***

Konnur Science PU College — Pre-University Course (PCMCs) — ***2022 – 2023*** | *Percentage:* ***95.16%***

EXPERIENCE

STON Technology — Software Development Engineer Intern — *Bengaluru, India | June 2026 – Sept 2026*

- Engineered an admit-card and attendance verification system utilizing QR-code scanning with automated Excel reporting pipelines.
- Implemented real-time gate check-in workflows, camera-based QR scanning, and walk-in candidate registration with MongoDB attendance persistence.
- Built secure student onboarding via CSV whitelisting, join-code account claiming, Google OAuth, Role-Based Access Control (RBAC), and TOTP two-factor authentication.
- Developed per-drive attendance analytics, officer verification attribution, automated formatting, and edge-case notification alerts for operational integrity.

FEATURED PROJECTS

PIHUB — Offline AI Tutor & Edge Distribution Platform (Flagship) — *github.com/Akash-S-G/pihub*

Technologies: Python, FastAPI, llama.cpp, GGUF Models, Qdrant, Docker, Raspberry Pi

- Built an offline-first learning platform enabling local AI tutoring, content pack distribution, and edge computing across classrooms.
- Architected FastAPI services running llama.cpp for quantized GGUF model execution, memory caching, and low-latency token streaming.
- Engineered a decentralized synchronization engine supporting 405+ curriculum content pack manifests, chunk offset resuming, and UDP beacon intranet discovery.

AI-Based Smart Shopping Cart System — *github.com/Akash-S-G/SMART-CART*

Technologies: FastAPI, Python, YOLO11n, PostgreSQL, WebSockets, Modal Cloud

- Created an automated checkout ecosystem integrating real-time visual product recognition with backend shopping cart and inventory state.
- Built vision detection endpoint using YOLO11n with cloud inference fallback, broadcasting instant item updates to carts via WebSockets.
- Implemented product catalog search, barcode generation, inventory decrement guards, coupon validation, and automated PDF billing.

Sol-Bridge — Community Solar Energy Sharing Platform — *github.com/Akash-S-G/Sol-Bridge*

Technologies: Node.js, Express.js, PostgreSQL, TimescaleDB, Redis, MQTT

- Developed backend microservices architecture for IoT smart-meter telemetry ingestion and community peer-to-peer energy sharing.
- Designed high-throughput time-series storage with TimescaleDB for continuous meter sensor metrics and Redis Pub/Sub for internal events.
- Structured user management, quota allocation workflows, and smart energy transaction verification boundaries.

FundShield — AI-Powered Welfare Verification Microservice — *github.com/Akash-S-G/FUNDSHIELD-ML*

Technologies: Python, FastAPI, OpenCV, YOLOv8, Scikit-learn, Docker

- Architected Dockerized ML verification service for construction proof validation and financial transaction anomaly detection.
- Implemented perceptual hashing (ImageHash) for duplicate detection, ExifRead for metadata verification, and YOLOv8 for stage classification.
- Deployed Scikit-learn Isolation Forest algorithm to surface objective anomaly risk scores without false conclusive fraud claims.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript
Backend Engineering: FastAPI, Flask, Express.js, REST APIs, WebSockets, Socket.IO
AI & Applied ML: LLM Integration, Local Inference (llama.cpp, GGUF), Computer Vision (OpenCV, YOLO), Scikit-learn
Databases & Caching: PostgreSQL, TimescaleDB, MongoDB, SQLite, Redis
Systems & DevOps: Linux, Raspberry Pi, Docker, Git, GitHub
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

ACHIEVEMENTS & PROBLEM SOLVING

- **DSA Problem Solving:** Solved **300+** data structures and algorithms problems across LeetCode and Codeforces.
- **Competitions:** Winner of the Technical Quiz Challenge at Nagarjuna College.